

PROJECT PROPOSAL

FAIRNESS IMPACT ON RAG PERFORMANCES

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Following the reading of the paper “Fairness and Transparency in ranking” (Castillo, 2018) which introduces the idea of fairness in retrieved documents rankings, we found a recent paper “Towards Fair RAG” (Eun Kim & Diaz, 2025) which extends the idea to a fairness in the reranking process of RAG.

One of the main conclusions of the paper is “RAG systems with fair rankings can maintain a high level of generation quality and, in many cases, even outperform traditional RAG systems”, which raises questions on our side, because based on literature, fairness definition and simple math, implementing ranking models with fairness methods introduces tradeoff between ensuring fairness and performances.

We’d like to test their conclusion and see whether it has unmentioned limitations, and whether the relationship between fairness and performance actually correlates as seen in their results (table 2)?

Also, we’d like to check a hypothesis we have – if fairness does improve performance in RAG systems, does it stem from simply increasing the diversity of the documents? (retrieving documents that are more different from each other is more beneficial for the generative model to answer questions more precisely), since Intuitively we’d guess that fairness in ranking inherently increases the diversity of retrieved documents.

For our research project, we propose to run similar experiments to the experiments in the paper “Towards fair RAG” (which means to use the same dataset – LaMP) while being critical to mistakes, possibly test additional metrics of performance and fairness to challenge the main conclusion, and additionally – measure diversity of documents to see if they correlate to fairness or to better performance.

Summary:

Literature - “Towards Fair RAG” (Eun Kim & Diaz, 2025), “Fairness and Transparency in ranking” (Castillo, 2018), “Evaluating Stochastic Rankings with Expected Exposure” (Diaz et al, 2020)



Research question – "Does Fairness directly improve RAG performance, or does the performance gain actually come from the increased diversity introduced through Fairness interventions?"

Dataset – LaMP benchmark dataset <https://lamp-benchmark.github.io/>